

SAVANNA

SENTINEL



Functional Requirements

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Functional Requirements

Dedicated Functional Requirements document can be found in the github docs folder

R1: User Access and Security Management

Subsystem: Auth Identity System

R1.1 Role Based Access Control (RBAC)

- **R1.1.1:** The system will support role-based access control with predefined roles including Ranger, Analyst, Community Liaison and Admin.
- **R1.1.2:** The system will enforce server-side authentication using stateless sessions (JWT).
- **R1.1.3:** The system will restrict access to specific views, features, and map layers based on the authenticated user's assigned role.

R1.2 Audit and Security

- **R1.2.1:** The system will maintain an audit log of all administrator-level actions, recording the actor's user ID, action performed, target record type and ID, and a timestamp
- **R1.2.2:** The audit log shall be readable only by Admins.
- **R1.2.3:** The audit log shall be immutable, no user, including Admins, shall be able to edit or delete an existing audit log entry.
- **R1.2.4:** The system will ensure all data in transit is encrypted via HTTPS.

R2: Data Ingestion and Validation

Subsystem: Data Ingestion Subsystem

R2.1: File Upload and Parsing

- **R2.1.1:** The system will allow Analysts to upload CSV files containing historical data.
- **R2.1.2:** The system will provide a data upload wizard that displays a preview of the uploaded data prior to commit.
- **R2.1.3:** The data upload wizard will display human-readable validation errors for any row that fails validation, without committing any data until the Analyst confirms the import.
- **R2.1.4:** The system will parse and translate uploaded data into core entities including: Incidents, Patrol Tracks, Sightings, and Tip-offs.

R2.2: Geospatial Storage

- **R2.2.1:** The system will persist validated geospatial data with spatial indexing to support efficient location-based querying.

R3: Geospatial Risk Engine and Visualization

Subsystem: Geospatial Risk Engine Subsystem

R3.1: Risk Engine Processing

- **R3.1.1:** The AI Risk Engine will process historical data via asynchronous background jobs to compute risk scores, such that the requesting user is not blocked while scores are calculated.
- **R3.1.2:** The Risk Engine will calculate risk hotspots gridded into geographic cells.
- **R3.1.3:** The Risk Engine will compute a distinct risk score per geographic cell for each defined time interval.

R3.2: Interactive Map

- **R3.2.1:** The system will display a heatmap layer on the map indicating high-risk areas based on the calculated risk scores.
- **R3.2.2:** The system will provide an interactive time range slider allowing users to select a time interval and view its corresponding risk hotspots
- **R3.2.3:** The system will provide interactive map controls including a legend describing the heatmap scale.
- **R3.2.4:** The system will provide interactive data filters allowing users to filter map layers.
- **R3.2.5:** The system will enforce role-based access control on all map data API endpoints, such that each role receives only the data it is authorised to access as defined by the RBAC policy.

R3.3 Location Tracking

- **R3.3.1:** The system will display an updating icon of the user's location on the map.
- **R3.3.2:** The system will track the user's location using the device's motion sensors if a GPS signal cannot be located.
- **R3.3.3:** The system will allow the user to select a known point of interest if their last known location cannot be determined.

R4: Risk Engine Explainability

Subsystem: Risk Explainability Subsystem

R4.1: Explainability Metrics

- **R4.1.1:** The Risk Engine will generate explainability metrics using SHAP for each computed risk score.
- **R4.1.2:** The generated explainability metrics will detail the contribution of each input feature to the resulting risk score.

R4.2: Explainability Panel

- **R4.2.1:** The system will provide an explainability UI panel that communicates the specific reasons a geographic cell is designated as high risk.
- **R4.2.2:** The explainability UI will display a confidence level for each risk score.

- **R4.2.3:** The explainability UI's confidence level display must not overclaim model certainty (e.g. must not present a score as a guarantee of an event occurring)

R5: Patrol Route Planning

Subsystem: Patrol Planning Subsystem

R5.1: Constraint-Based Route Generation

- **R5.1.1:** The Patrol Planner will accept custom user parameters, including starting point.
- **R5.1.2:** The Patrol Planner will consume the current Risk Heatmap and the supplied resource constraints to generate multiple alternative patrol routes per planning request.
- **R5.1.3:** All route alternatives generated from a single planning request will share a common request ID and will be returned together to the requesting user.

R5.2: Route Retrieval

- **R5.2.1:** The system will allow authenticated Rangers and Analysts to retrieve their previously generated patrol route requests.
- **R5.2.2:** Retrieved route alternatives will be grouped by their shared request ID.

R5.3: Route Evaluation

- **R5.3.1:** The system will provide a side-by-side route comparison interface allowing rangers to evaluate proposed routes.
- **R5.3.2:** The route comparison interface will display, for each route option, the estimated risk coverage, estimated time and estimated fuel consumption.

R6: Field Reporting and Offline Sync

Subsystem: Field Reporting Subsystem

R6.1: Mobile Offline Capture

- **R6.1.1:** The system will provide a Progressive Web App (PWA) interface for rangers to capture Field Reports.
- **R6.1.2:** The system will allow Rangers to input field report data locally on the device when operating without network connectivity.
- **R6.1.3:** A Ranger may submit multiple field reports during a single patrol.
- **R6.1.4:** Each field report shall be classified as either an incident report or a sighting report.
- **R6.1.5:** An incident report shall capture an incident type and a severity level, in addition to the common fields.
- **R6.1.6:** A sighting report shall capture a species name and an optional animal count, in addition to the common fields.
- **R6.1.7:** Each submitted field report will create a corresponding geospatial event of the matching type (Incident or Sighting).
- **R6.1.8:** Offline field report data shall be persisted to the device using IndexedDB until synchronisation occurs.
- **R6.1.9:** The system will optionally support capturing photo evidence associated with a field report.
- **R6.1.10:** The system will upload captured photo evidence to S3-compatible storage (Seaweed) via pre-signed URLs.

R6.2: Field Report Management

- **R6.2.1:** The system will allow Rangers to edit their own previously submitted field reports, both online and offline.
- **R6.2.2:** For all field reports, the editable common fields are description, location, occurred_at and associated photos.
- **R6.2.3:** For incident reports specifically, incident type and severity shall additionally be editable.
- **R6.2.4:** For sighting reports specifically, species and count shall additionally be editable.
- **R6.2.5:** The system will allow Rangers to delete their own previously submitted field reports.
- **R6.2.6:** Deleted field reports will be soft-deleted and retained in the database for audit purposes.
- **R6.2.7:** The system will allow Admins to edit any field report regardless of the submitting Ranger, with the same field-level permissions defined in R6.2.2–R6.2.4.
- **R6.2.8:** The system will allow Admins to delete any field report regardless of the submitting Ranger.
- **R6.2.9:** The system will allow Users to leave comments on reports.
- **R6.2.10:** The system will allow Users to change the status of reports from Unknown, Unresolved, and Resolved.

R6.3: Synchronisation and Conflict Handling

- **R6.3.1:** The system will allow Admins to delete any field report regardless of the submitting Ranger.
- **R6.3.2:** The Sync Service will detect database conflicts arising from offline data uploads and offline deletions.
- **R6.3.3:** The Sync Service will resolve detected conflicts deterministically, such that no record is silently overwritten.
- **R6.3.4:** Soft-deleted records created offline shall be correctly propagated to the central database upon sync.

R7: Community Tip-off Submission

Subsystem: Community Tip-off Subsystem

R7.1: Tip-off Capture

- **R7.1.1:** The system will allow Community Liaisons to submit tip-offs describing an observed incident or sighting.
- **R7.1.2:** Every tip-off shall capture report type, location, occurred_at and description.
- **R7.1.3:** The system will optionally allow photo evidence to be attached to a tip-off.
- **R7.1.4:** An incident tip-off shall additionally capture an incident type and a severity level.
- **R7.1.5:** A sighting tip-off shall additionally capture a species name and an optional animal count.
- **R7.1.6:** The system will reject a tip-off submission if the occurred_at field is set to a future date or time.
- **R7.1.7:** The system will associate each submitted tip-off with the submitting Community Liaison's user account.

R8: Dashboards and System Analytics

Subsystem: Analytics & Dashboard Subsystem

R8.1: Operational Dashboards

- **R8.1.1:** The system will display dashboard cards summarising patrol coverage
- **R8.1.2:** The system will display dashboard cards summarising Field Report findings.
- **R8.1.3:** The system will display dashboard cards summarising operational trends over time.
- **R8.1.4:** The system will display basic model performance metrics to Analysts.
- **R8.1.5:** Displayed model performance metrics shall be clearly labelled and must not overclaim model certainty.

R9: Account and Identity Management

Subsystem: Account & Identity Management Subsystem

R9.1: Authentication Flow

- **R9.1.1:** The system will allow only activated user accounts to log in.
- **R9.1.2:** The system will deny login access to accounts pending Admin activation.
- **R9.1.3:** The system will return a deliberately vague error response for any failed login attempt, such that the response does not reveal whether the email address exists, the password is incorrect or the account is inactive.
- **R9.1.4:** The system will allow users to log out, securely invalidating their current active session.
- **R9.1.5:** The system will allow clients to obtain a new access token using a valid, non-revoked refresh token without requiring the user to log in again.
- **R9.1.6:** The system will return a 401 Unauthorised response if a client attempts to use an invalid or revoked refresh token.
- **R9.1.7:** The system will provide a password recovery mechanism whereby a user submits their registered email address and the system sends a time-limited, single-use password reset link to that address.
- **R9.1.8:** Following a valid password reset link will allow the user to set a new password without providing their current one.
- **R9.1.9:** The system will invalidate a password reset token after its first use or after its expiry, whichever occurs first.
- **R9.1.10:** The system will request a pin to be entered that will be sent to the user's email when the account is identified to have admin role permissions.

R9.2: User Registration

- **R9.2.1:** The system will allow prospective users to register for an account by providing a username, email address, password, first name and last name.
- **R9.2.2:** At registration, the system will require the prospective user to select their intended role from the available options: Ranger, Analyst, or Community Liaison.
- **R9.2.3:** All newly registered accounts will be created in an inactive state.
- **R9.2.4:** The system will require explicit activation by an Admin before a newly registered user can log in.

R9.3: User Account Administration

- **R9.3.1:** The system will allow an Admin to view a comprehensive list of all system users and their current statuses, including pending activation requests.
- **R9.3.2:** The system will allow an Admin to activate a user account.
- **R9.3.3:** The system will allow an Admin to deactivate a user account.
- **R9.3.4:** The system will allow an Admin to modify the role of any non-Admin user account.
- **R9.3.5:** The system will prevent any Admin from activating, deactivating, or modifying the role of an account that holds the Admin role.
- **R9.3.6:** The system will allow an Admin to permanently delete a user account that is in an inactive (pending) state.
- **R9.3.7:** The system will prevent the deletion of any user account that is in an active state, an Admin must first deactivate an account before deletion is permitted.
- **R9.3.8:** Upon deletion, the system will permanently remove the user record and all associated tokens (refresh tokens, password reset tokens) from the database, with no soft-delete applied.
- **R9.3.9:** The system will prevent any Admin from deleting an account that holds the Admin role.
- **R9.3.10:** The system will record a user account deletion in the audit log, including actor ID, action, target user ID, and timestamp, before the record is removed.

R9.4: Profile Management

- **R9.4.1:** The system will allow users to view their own basic profile information.
- **R9.4.2:** The system will allow users to update their own basic profile information.
- **R9.4.3:** The system will allow users to securely update their own password.
- **R9.4.4:** A password change request shall require the user's current password for verification before the change is accepted.
- **R9.4.5:** On a successful password change, the system will revoke all existing refresh tokens associated with that account.

R10: General Application Features

Subsystem: Core Application Services Subsystem

R10.1: Data Entry and Validation

- **R10.1.1:** The system will enforce client-side form validation on all user inputs.
- **R10.1.2:** The system will enforce server-side form validation on all user inputs, independent of client-side validation, to prevent malformed data
- **R10.1.3:** The system will require user passwords to be a minimum of 8 characters in length.
- **R10.1.4:** The system will display clear, human-readable error messages when form validation fails.

R11: Geospatial Workspace

Subsystem: Geospatial Workspace

R11.1: Drawing System

- **R11.1.1:** The system will allow the user to draw on the map, using Polygons, Line, Freedraw, and Point objects.
- **R11.1.2:** The system will allow the user to undo changes on newly added map elements.

R11.2: Map Elements

- **R11.2.1:** The system will allow the user to annotate map elements, including assigning labels, icons, and colours.
- **R11.2.2:** The system will allow a buffer to be assigned to an element to display the effect it has on the surrounding area.
- **R11.2.3:** The system will allow rules to be assigned to a map element.

R11.3 Rule system

- **R11.3.1:** The system will allow rules to be assigned to map elements. to increase/decrease the risk associated with the area, and/or reward or punish the patrol planner for following the element
- **R11.3.2:** The system will allow priorities to be assigned to elements, which will take precedence over any intersecting elements with a lower priority.
- **R11.3.3:** The system will allow a buffer decay to be specified for an element, decreasing the strength of the associated rule with distance from the element.
- **R11.3.4:** The system will allow the rules of an element to be toggled on or off without needing to modify the strength of the associated rule.

R11.4 Layer System

- **R11.4.1:** The system will allow the user to create layers, and create elements/layers as children of a layer.
- **R11.4.2:** The system will allow the user to toggle the visibility of layers/elements
- **R11.4.3:** The system will allow the user to move elements between layers.
- **R11.4.4:** The system will allow the user to delete elements/layers.